

Solving The Consumer's Problem in a Dynamic Model with Government

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I Introduction

In this dynamic model with an infinite time horizon, the government purchases consumption goods g_t in period $t = 0, 1, 2, 3, \dots$ and is financed through τ_t units of lump sum taxes. Also, the government issues one period bonds b_t (in period $t - 1$) where principal and interest on the debt issued in the previous period is $(1 + r_t)b_t$ where r_t is the interest rate. Note that we assume $b_0 = 0$ and that there is no investment (and therefore no capital) for simplicity and to drive intuition before adding it in. So, the representative firm's production technology can be given by $y_t = z_t n_t$. Where y_t is output in period t , z_t is total factor productivity in period t , and n_t is the quantity of labor in period t . Finally, in each period, the consumer is endowed with one unit of time, for example 24 hours.

II The Consumer's Problem

Since this model has an infinite horizon economy, the consumers utility function can be given by

$$\sum_{t=0}^{\infty} \beta^t u(c_t, \ell_t) \quad (1)$$

Where c_t is the quantity of consumption goods in period t , ℓ_t is the quantity of leisure in period t , w_t is real wage in period t , and $0 < \beta < 1$ is the discount factor¹. Letting δ be the discount rate, we obtain $\beta = \frac{1}{1 + \delta}$ where $\delta > 0$. Then, the consumer's problem is

$$\max_{\{c_t, s_{t+1}, \ell_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t, \ell_t)$$

Subject to

$$c_t + s_{t+1} = w_t(1 - \ell_t) - \tau_t + (1 + r_t)s_t \quad (2)$$

[†]Definitions and example problems from ECON 62051: Macroeconomic Theory I instructed by Dr. Duhyeong Kim and *Notes on Macroeconomic Theory* by Steve Williamson. Kent State University.

¹A proof for the discount factor can be found in *Games of Strategy* by Dixit et al.

Where s_{t+1} is the amount of bonds purchased by the consumer in period t which will be due in period $t + 1$. We can assume that there is no infinite borrowing (AKA Ponzi Schemes) so that in equilibrium,

$$\lim_{t \rightarrow \infty} \frac{s_{t+1}}{\prod_{i=1}^t (1 + r_i)} = 0 \quad (3)$$

The above equation implies that we can define an intertemporal budget constraint such that the consumers problem becomes

$$\max_{\{c_t, s_{t+1}, \ell_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t, \ell_t)$$

Subject to

$$c_0 + \frac{c_t}{\prod_{i=1}^t (1 + r_i)} = w_0(1 - \ell_0) - \tau_0 + \sum_{t=1}^{\infty} \frac{w_t(1 - \ell_t) - \tau_t}{\prod_{i=1}^t (1 + r_i)} \quad (4)$$

We can derive Equation 4 by using repeated substitution. Start with the consumers budget constraint in period 0 and 1 respectively,

$$\begin{aligned} c_0 + s_1 &= w_0(1 - \ell_0) - \tau_0 + (1 + r_0)s_0 \\ c_1 + s_2 &= w_1(1 - \ell_1) - \tau_1 + (1 + r_1)s_1 \end{aligned}$$

Then, we solve for $s_1 = \frac{1}{1+r_1}(c_1 + s_2 - w_1(1 - \ell_1) + \tau_1)$ and substitute it into our equation for period zero. Note that $s_0 = 0$. Using repeated substitution and taking the limit of the s_{t+1} term,

$$c_0 + \frac{c_1}{\prod_{i=1}^1 (1 + r_i)} + \frac{c_2}{\prod_{i=1}^2 (1 + r_i)} + \dots + \lim_{t \rightarrow \infty} \frac{s_{t+1}}{\prod_{i=1}^t (1 + r_i)} = w_0(1 - \ell_0) - \tau_0 + \sum_{t=1}^{\infty} \frac{w_t(1 - \ell_t) - \tau_t}{\prod_{i=1}^t (1 + r_i)}$$

Since we assume no ponzi scheme, the limit converges to zero, giving Equation 4. Now we are ready to obtain our first order conditions. Let the Lagrangian function be

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t u(c_t, \ell_t) - \lambda \left(w_0(1 - \ell_0) - \tau_0 + \sum_{t=1}^{\infty} \frac{w_t(1 - \ell_t) - \tau_t}{\prod_{i=1}^t (1 + r_i)} - c_0 - \frac{c_t}{\prod_{i=1}^t (1 + r_i)} \right)$$

Deriving the first order conditions,

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial c_t} &= \beta^t u_1(c_t, \ell_t) - \lambda \cdot \frac{1}{\prod_{i=1}^t (1 + r_i)} = 0 \\ \frac{\partial \mathcal{L}}{\partial \ell_t} &= \beta^t u_2(c_t, \ell_t) - \lambda \cdot \frac{w_t}{\prod_{i=1}^t (1 + r_i)} = 0 \end{aligned}$$

Note that

$$\begin{aligned} \beta^t u_1(c_t, \ell_t) &= \lambda \cdot \frac{1}{\prod_{i=1}^t (1 + r_i)} \\ \beta^t u_2(c_t, \ell_t) &= \lambda \cdot \frac{w_t}{\prod_{i=1}^t (1 + r_i)} \end{aligned}$$

so

$$\frac{u_2(c_t, \ell_t)}{u_1(c_t, \ell_t)} = w_t \quad (5)$$

Which is the $MRS_{\ell,c}$ or the price of leisure relative to consumption goods, where u_1 denotes the partial derivative of the utility function with respect to c_t and u_2 with respect to ℓ_t . Finally, we can find the price of current goods relative to future goods,

$$\frac{\partial \mathcal{L}}{\partial c_{t+1}} = \beta^{t+1} u_1(c_{t+1}, \ell_{t+1}) - \lambda \cdot \frac{1}{\prod_{i=1}^{t+1} (1 + r_i)} = 0$$

Note that

$$\frac{\lambda}{\prod_{i=1}^{t+1} (1 + r_i)} = \frac{\lambda}{\prod_{i=1}^t (1 + r_i)} \cdot \frac{1}{(1 + r_{t+1})}$$

so

$$\frac{\partial \mathcal{L}}{\partial c_{t+1}} = \beta^{t+1} u_1(c_{t+1}, \ell_{t+1}) - \frac{\frac{\lambda}{\prod_{i=1}^t (1 + r_i)}}{(1 + r_{t+1})} = 0$$

Where the numerator of the fraction is equal to $\beta^t u_1(c_t, \ell_t)$, so,

$$\beta^{t+1} u_1(c_{t+1}, \ell_{t+1}) \frac{\beta^t u_1(c_t, \ell_t)}{(1 + r_{t+1})} = 0$$

Which implies that

$$\beta u_1(c_{t+1}, \ell_{t+1}) = \frac{u_1(c_t, \ell_t)}{(1 + r_{t+1})} = 0$$

which allows us to arrive at our final equation

$$(1 + r_{t+1}) = \frac{u_1(c_t, \ell_t)}{\beta u_1(c_{t+1}, \ell_{t+1})} \quad (6)$$

Which is equal to $MRS_{c_t, c_{t+1}}$